

Sanatbek Zhamurzaev

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- Quantitative Researcher and Data Engineer with a strong foundation in statistical modeling, time-series analysis, and algorithmic data structures. Experienced in building robust ETL pipelines, processing complex financial datasets, and developing automated trading signals. Demonstrated ability to optimize computational algorithms and extract actionable anomalies from large-scale data.

Education

Eötvös Loránd University (ELTE)

Bachelor of Science in Computer Science (Stipendium Hungaricum Scholar, Top 5%)

September 2023 – Present

Budapest, Hungary

Experience

Research Assistant (Data Analysis)

May 2025 – October 2025

Under Dr. Alexander Chervov

Remote

- Engineered robust data pipelines for cleaning, transforming, and visualizing 10+ GB of complex multi-dimensional datasets using Python, Pandas, and NumPy.
- Applied statistical methods to extract and validate statistically significant anomalies within large-scale structured data, co-authoring 50+ pages of quantitative research.

Teaching Assistant – Algorithms & Data Structures

January 2026 – June 2026

Eötvös Loránd University (ELTE)

Budapest, Hungary

- Evaluated rigorous algorithmic implementations for 300+ students, strictly analyzing time/space computational complexity (Big-O), memory allocation, and optimal sorting strategies.
- Graded 500+ assignments per semester, focusing on edge-case handling and graph algorithms, improving average student code efficiency metrics by 25%.

Data Pipeline Engineer

November 2025 – June 2026

ToiWare

Remote

- Architected high-throughput ETL pipelines integrating asynchronous APIs, optimizing data flow and parsing speed for backend systems.
- Developed Python modules featuring multi-key deduplication and automated data validation, reducing formatting errors by 35% and ensuring strict data integrity for downstream analytics.

Quantitative & Engineering Projects

Algorithmic Trading Signals (Retail Crypto & TradFi)

Python | Pandas | Time-Series

- Developed automated Python scanners for retail markets, analyzing 5-minute OHLCV data to detect structural anomalies correlated with Open Interest spikes and volume divergences.
- Implemented a Japanese candlestick pattern recognition engine detecting 15+ patterns (e.g., engulfing, doji, hammer) with configurable statistical sensitivity thresholds.

LauzHack (UBS) — Financial Anomaly Detection

Python | Scikit-learn | Statistical Methods

- Architected a scalable anomaly detection engine under a strict 24-hour hackathon deadline.
- Applied statistical feature engineering and outlier detection models (z-score filtering, density estimation) to process and classify 10,000+ rows of transactional financial data.

ClearMind — Vector Retrieval Architecture

Python | Vector Search | FastAPI

- Engineered a low-latency data retrieval backend handling 100+ daily queries utilizing vector similarity indexing.
- Optimized the retrieval pipeline and API routing, reducing search latency by 30% through efficient embedding caching and index optimization.

Technical Skills

- **Programming Languages:** Python, C, Java, SQL, Bash.

- **Quantitative & Data:** Time-Series Analysis, Algorithmic Signal Generation, Candlestick Pattern Recognition, Statistical Modeling, Pandas, NumPy, SciPy, Scikit-learn, EDA.

- **Engineering:** Data Structures, Algorithmic Complexity (Big-O), ETL Pipelines, FastAPI, Vector Databases.

- **Tools & Infrastructure:** Git/GitHub, Linux, Docker, Jupyter Notebooks, LaTeX.

- **Languages:** English (Fluent), Kyrgyz (Native), Russian (Native), Hungarian (Beginner).